

Figure 13: THS off, low memory compaction contiguity CDF.

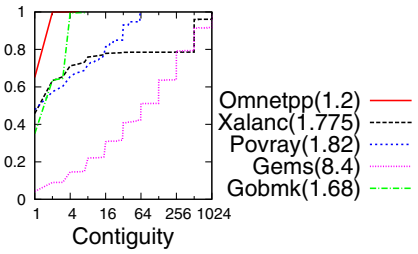


Figure 14: THS off, low memory compaction contiguity CDF.

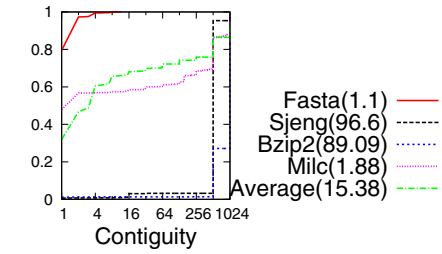


Figure 15: THS off, low memory compaction contiguity CDF.

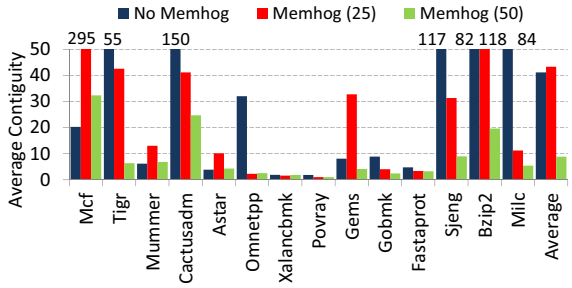


Figure 16: Average contiguity for THS on, normal memory compaction with varying `Memhog`.

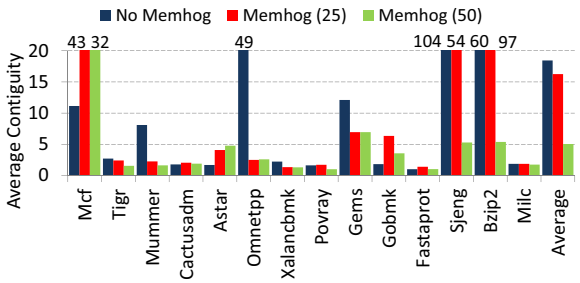


Figure 17: Average contiguity for THS off, normal memory compaction with varying `Memhog`.

the same hit time. After recording these misses, we then run the same benchmarks on configurations with CoLT-SA, CoLT-FA, and CoLT-All, tracking the new TLB miss rates. We assume that CoLT-SA uses VPN[4-2] and VPN[6-2] for L1 and L2 set selection, meaning that up to four translations can be coalesced per entry (we will later show the effect of using more aggressive indexing). We also conservatively assume 8-entry fully-associative TLBs when using CoLT-FA and CoLT-All.

First and foremost, Figure 18 shows that all three CoLT schemes improve *every* single benchmark by eliminating large chunks of the baseline misses. On average, CoLT-SA eliminates 40% of both L1 and L2 TLBs misses, while CoLT-FA and CoLT-All do even better, eliminating around 55% of both L1 and L2 misses. Second, Figure 18 shows that many of the benchmarks experiencing TLB pressure gain particularly from CoLT. For example, `Mcf`, `CactusADM`, and `Astar` all eliminate vast amounts of their TLB misses. In fact, `Astar` almost achieves perfect TLBs with no misses with CoLT-FA and CoLT-All.

Third, there is a correlation between system contiguity and effectiveness of CoLT. For example, `Mcf`, `Bzip2`, `Milc`, and `CactusADM`, which all see more instances of 20-page contiguity on average, can coalesce large amounts of translations, increasing TLB reach substantially. However, contiguity alone does not guarantee coalescing success; for coalescing to be effective, contiguous entries must actually be used close together in time. Without this temporal proximity, a coalesced entry will be evicted from the TLB before multiple member translations are used. This explains the lower benefits of `Tigr`, which sees 10% TLB miss elimination rates despite a contiguity of over 50 pages on average.

Fourth, Figure 18 shows that leveraging the superpage TLB in CoLT-FA and CoLT-All provides 10-15% gains over CoLT-SA on average. Benchmarks like `Mcf` and `Fastaprot` benefit particularly from this. These gains are achieved *despite* dropping from a 16-entry to an 8-entry structure. We find that the primary reason for this is that even with THS on, superpages are used sparingly. Therefore, a surprisingly high number of entries remain wasted in the fully-associative TLB in the baseline case. Instead, CoLT-FA and CoLT-All use these entries and can even perform unrestricted coalescing on them, unlike the set-associative TLBs.

The difference between CoLT-FA and CoLT-All remains more nuanced. We find generally that both schemes eliminate roughly 55% of TLB misses on average. Generally on the more TLB-intensive benchmarks (eg. `Mcf`, `Tigr`, `Mummer`, `CactusADM`), CoLT-All outperforms CoLT-FA slightly. However, in many benchmarks, CoLT-All falls surprisingly short of CoLT-FA. This occurs because CoLT-FA is better able to coalesce translations that reside across multiple LLC cache lines. Essentially, in these benchmarks only a few translations in a single cache line are coalescible with translations from another cache line. In CoLT-FA, since all the translations are brought into the fully-associative structure, these entries are coalesced. In CoLT-All however, if the cache line that maintains only a few coalescible translations falls below the pre-defined threshold, they are inserted into the set-associative TLB and can therefore never be merged with the second cache line's translations (which sits in the fully-associative TLB). This reduces CoLT-All's hit rates compared to CoLT-FA.

Overall, all CoLT designs eliminate a large fraction of TLB misses. We now focus on implementation details of the various CoLT designs to lend greater insight on our gains.

7.1.2. Impact of CoLT-SA's indexing scheme on TLB miss rates. Our initial CoLT-SA results assume that we use VPN[4-2] and VPN[6-2] for L1 and L2 set selection. This limits the amount of coalescing to four translations per entry. While additional contiguity could be coalesced by further left-shifting the index bits, this also increases conflict misses. Figure 19 studies these opposing forces on the 4-way associative TLBs by left-shifting the traditional index bits by one bit (VPN[3-1] and VPN[5-1] for L1 and L2 TLBs), two bits, and three bits (VPN[5-3] and VPN[7-3] for L1 and L2 TLBs). These correspond to maximum allowable coalescing of two, four and eight translations.